



Luca Domeniconi

AI Researcher graduated in Artificial Intelligence at the University of Bologna

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Education

M.Sc. University of Bologna , Artificial Intelligence - Thesis: <i>Using Sparse Features to Characterize and Mitigate Concept-Dependent Modality Gap in Vision-Language Models</i> - Grade: 110/110 with honors	Bologna, Italy Sept 2023 – Mar 2026
B.Sc. University of Bologna , Computer Engineering Grade: 109/110	Bologna, Italy Sept 2020 – Oct 2023

Publications

How to Evaluate and Refine your CAM Luca Domeniconi, Alessandra Stramiglio, Michele Lombardi, Samuele Salti (ICPR 2026)	Publication Repository
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Experience

EPFL , Research Intern (Master's Thesis) Conducted research on interpretability and vision-language models for my Master's thesis project in MINTS Lab, under the supervision of Prof. Andrea Cavallaro. - Studied the semantical structure of the modality gap inside CLIP's embedding space - Proposed to use Sparse Autoencoders to better characterize the modality gap	Lausanne, Switzerland Oct 2025 – Mar 2026 6 months
Belluzzi Fioravanti High School , Computer Science Teacher Taught students about computer architecture and networking using sockets	Bologna, Italy Jan 2024 – Feb 2024 2 months
Gruppo Finmatica , Software Engineer Intern Deployment of a web based collaborative document editor and development of a React Native prototype for barcode scanning to assist warehouse personnel in inventory management	Bologna, Italy Mar 2023 – June 2023 4 months
Private Tutor Private computer science and mathematics lessons for undergraduate and high school students	Rimini, Italy 2019 – 2025 6 years

Achievements

1st Place at EPFL LauzHack 2024, BMS Challenge Developed an explainable time-series forecasting model using Gaussian Processes and Shapley values to predict pharmaceutical demand, ranking 1st out of 12 teams in the Bristol Myers Squibb 24-hour hackathon challenge.	Lausanne, Switzerland
1st Place at Tablut AI Agent Challenge Designed and developed an AI player for the asymmetric board game Tablut using adversarial search and heuristic evaluation, ranking 1st out of 10 agents.	University of Bologna

Selected Projects

ICD Multimodal Diagnosis Implemented a Tree-constrained multimodal ICD-9 generation pipeline from chest X-rays and clinical text using custom made VLMs	Project Repository
Semantic Segmentation of Unexpected Objects on Roads Implemented an open-world semantic segmentation model using a pre-trained DINOv3 plus metric learning to segment known classes while detecting anomalous regions.	Project Repository

Constraint Programming for Multiple Couriers Optimization

Modelled the Capacitated Vehicle Routing Problem using CP, SAT, SMT, and MILP (MiniZinc, OR-Tools, Z3, Gurobi)

[Project Repository](#) ↗

Shelves Product Detection

Instance detection of products on shelves using only classical computer vision techniques with OpenCV

[Project Repository](#) ↗

Skills

Research Interests: Explainability, Attribution Methods, Vision-Language Models

Computer Vision: Python, OpenCV, PyTorch

Machine Learning: Keras, numpy, pandas, scikit-learn, Matplotlib

Web and Mobile Development: JavaScript, NodeJS, React, React Native, Android Native

Programming Languages and Tools: Java, C, C#, LaTeX, SQL, MATLAB, Minizinc, Unity 3D, Docker, Blender